

Digital Screen Network Management Platform

User Roles & Functionalities Specification

Purpose of this document

- [] This document defines the different user roles and their associated functionalities for a digital screen network management platform. It is intended to serve as a functional reference for software designers and developers.

1. Platform Overview

- [] The platform is designed primarily for the retail industry and enables Softomedia and its retail partners to jointly operate a network of digital screens deployed across retail locations.
- [] The network is co-owned by Softomedia and the retailer:
 - [] Softomedia operates the platform, commercializes advertising inventory, orchestrates campaigns, and ensures technical reliability.
 - [] Retailers co-own the network, host the screens in their stores, validate all content prior to broadcast, and may run their own campaigns.
- [] The broadcasting model is based on a fixed-loop scheduling logic, optimized for retail environments.
- [] This document focuses on the prototype (MVP) required to demonstrate core value and operational feasibility.

2. User Roles Overview

- [] The platform supports the following MVP user roles:
 - [] Super Administrator (Softomedia – Platform Owner)
 - [] Retailer Administrator (Retail Partner)
 - [] Softomedia Content & Campaign Manager
 - [] Advertiser / Agency User
 - [] Technical Operator (In-house CTO / Ops)
- [] Roles such as advanced analysts or external viewers are considered post-MVP.

3. User Roles & Functionalities

3.1 Super Administrator (Softomedia – Platform Owner)

Description:

- ☐ Softomedia internal role with full control over the platform, network, monetization, and governance.

Core Functionalities (MVP):

- ☐ Create and manage retailer accounts
- ☐ Create and manage advertiser / agency accounts
- ☐ Full user & role management
- ☐ Global platform configuration
- ☐ Screen network ownership and governance
- ☐ Campaign prioritization rules (paid vs retailer vs internal)
- ☐ Monetization rules (inventory allocation, pricing logic – high level)
- ☐ Global analytics and performance dashboards
- ☐ Audit logs and security oversight

3.2 Retailer Administrator (Retail Partner)

Description:

- ☐ Represents a retail organization hosting Softomedia screens in its locations.

Core Functionalities (MVP):

- ☐ Manage retail locations and store profiles
- ☐ View screens installed in their locations
- ☐ Validate, approve, or reject advertising content before broadcast
- ☐ Define local content constraints (categories allowed / excluded)
- ☐ Launch optional retailer-owned campaigns (store promotions)
- ☐ View performance metrics limited to their locations
- ☐ Communicate feedback or issues to Softomedia

3.3 Softomedia Content & Campaign Manager

Description:

- ☐ Softomedia internal role responsible for operational execution of campaigns and content.

Core Functionalities (MVP):

- ☐ Upload and manage media assets (ads, promos, HTML5)
- ☐ Create advertiser and retailer campaigns
- ☐ Build playlists and assign inventory
- ☐ Schedule campaigns by retailer, location, screen group, and time
- ☐ Submit campaigns for retailer validation
- ☐ Monitor campaign delivery and playback
- ☐ Manage content lifecycle (active / paused / expired)

3.4 Advertiser / Agency User

Description:

- ☐ Brands or agencies purchasing ad space from Softomedia.

Core Functionalities (MVP):

- ☐ Upload ad creatives
- ☐ Create campaign requests (budget, duration, targeting)
- ☐ Select preferred retail locations (subject to availability)
- ☐ Track campaign status (pending approval, live, completed)
- ☐ View basic campaign performance metrics
- ☐ Access invoices and campaign summaries

3.5 Technical Operator (In-house CTO / Ops)

Description:

- ☐ Softomedia technical role managing installation, reliability, and platform stability.

Core Functionalities (MVP):

- ☐ Register and provision new screens
- ☐ Assign screens to retailers and locations
- ☐ Monitor device health (online/offline, last heartbeat)
- ☐ Remote restart and basic diagnostics
- ☐ Software updates and configuration
- ☐ Incident tracking and resolution

4. Core MVP Functional Modules

4.1 Broadcasting & Scheduling Engine (Core MVP)

- ☐ Broadcasting is based on hourly loops
- ☐ Each loop contains 12 ads
- ☐ Each ad has a fixed duration of 5 seconds
- ☐ Total loop duration: 60 seconds
- ☐ The loop configuration changes every hour

Scheduling Rules:

- ☐ Advertisers must submit creatives strictly in 5-second format (video or static image)
- ☐ No variable ad duration in MVP
- ☐ Each hour has a predefined loop (playlist)
- ☐ Loops are generated one day in advance (D-1)

4.2 Campaign & Inventory Management

- ☐ Centralized inventory management
- ☐ Inventory allocation per loop and per hour
- ☐ Distinction between:
 - ☐ Paid advertiser campaigns
 - ☐ Retailer-owned campaigns
- ☐ Campaign priority rules defined by Softomedia

4.3 Retailer Validation Workflow (Mandatory)

- ☐ Daily schedules (hourly loops) are generated the day before broadcast
- ☐ Retailers receive a schedule preview per location
- ☐ Retailers can:
 - ☐ Approve the full schedule
 - ☐ Reject specific ads
 - ☐ Request replacements
- ☐ Only approved schedules are eligible for broadcast
- ☐ Approval status is logged and auditable

4.4 Content Specifications & Compliance

- ☐ Accepted formats: MP4 (video), JPG/PNG (image)
- ☐ Fixed duration: 5 seconds
- ☐ Resolution and aspect ratio enforced per screen type
- ☐ Automatic rejection if specifications are not met

4.5 Device & Network Monitoring

- ☐ Real-time screen status
- ☐ Proof-of-play logging
- ☐ Offline fallback loop
- ☐ Alerting for failures

4.6 Analytics (MVP)

- ☐ Proof-of-play per ad
- ☐ Loop delivery confirmation
- ☐ Hourly broadcast logs
- ☐ Campaign-level summaries

3.7 Viewer / Read-only Analyst (Optional)

Description:

- ☐ Has access to dashboards and reports without modification rights.

Core Functionalities:

- ☐ View dashboards and KPIs
- ☐ Export reports
- ☐ Access historical performance data
- ☐ No content or configuration modification rights

4. Cross-Role Functional Modules

- ☐ These modules may be accessed differently depending on role permissions:
 - ☐ Content Management System (CMS)
 - ☐ Scheduling & Playback Engine
 - ☐ Device Management & Monitoring
 - ☐ Analytics & Reporting

- ☐ Monetization & Billing
- ☐ Workflow & Approval Engine
- ☐ Security, Authentication & Audit Logs

5. Non-Functional Considerations (for Developers)

- ☐ Role-Based Access Control (RBAC)
- ☐ Multi-tenant architecture
- ☐ Scalability for thousands of screens
- ☐ High availability & offline fallback
- ☐ Security (encryption, access logs)
- ☐ API-first design

6. Next Steps (Optional Enhancements)

- ☐ Audience measurement & sensors integration
- ☐ Programmatic advertising
- ☐ AI-driven content optimization
- ☐ POS and retail media integrations